B.Tech. Degree III Semester Examination, November 2009

ME 303 THERMAL ENGINEERING I
(1999 Scheme)

Time: 3 Hours — Maximum Marks: 100

(All questions carry ***EQUAL*** marks)

I. (a) Explain Second Law of Thermodynamics.
(b) Calculate the available energy in 40 kg of water at 75 °C with respect to the surroundings at 5 °C, the pressure of water being 1 atmosphere.

OR

II. Determine the adiabatic flame temperature when liquid octane at 25 °C is burned with 300% theoretical air at 25 °C in a steady flow process.

III. (a) Distinguish between 2-stroke and 4-stroke engines.
(b) Explain in detail about knocking.

OR

IV. (a) Describe the stages of combustion in petrol engines.
(b) Explain the steps in drawing a heat balance sheet for JC engines.

V. (a) Explain the working of a roots blower with neat sketch.
(b) List out the advantages of multistage compression over single stage compression.

OR

VI. A single-acting two-stage air compressor deals with 4 m^3/minute of air at 1.013 bar and 15 °C with a speed of 250 rpm. The delivery pressure is 80 bar. Assuming complete intercooling, find the minimum power required by the compressor and the bore and stroke of the compressor. Assume a piston speed of 3m/sec, mechanical efficiency of 75% and volumetric efficiency of 80% per stage. Assume the polytropic index of compression in both the stages as 1.25 and neglect clearance.

VII. (a) Derive general heat conduction equation in Cartesian co-ordinates.
(b) The inner surface of a plane brick wall is at 50°C and the outer surface is at 30° C. Calculate the rate of heat transfer per m^2 of surface area of the wall, which is 250 mm thick. The thermal conductivity of the brick is 0.52 W/m °C.

OR

VIII. Determine heat lost by radiation per metre length of 80 mm diameter pipe at 300 °C, if
(a) located in a large room with red brick walls at a temperature of 27 °C
(b) enclosed in a 160 mm diameter red brick conduit at a temperature of 27 °C. Take emissivities of pipe and brick conduit as 0.79 and 0.93 respectively.

IX. (a) Explain the significance of following dimensionless numbers:
(i) Reynold's number
(ii) Prandtl number
(b) Derive an expression for the LMTD of a parallel flow heat exchanger.

OR

X. Water at the rate of 80 kg/minute is heated from 40°C to 70 °C by an oil having a specific heat of 1.88 KJ/kg k. The fluids are used in counter flow and the oil enters the heat exchanger area at 110 °C and leaves at 60 °C. Calculate the heat exchanger area assuming the overall heat transfer coefficient of 349 W/m^2k.

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